Physics Isomorphic Questions

Multiple-Choice Question Set

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. Which of the following statements describes a reasonable physics idealization?

A. A bicycle wheel can be treated as a point mass when analyzing how different points on the rim move during rotation.
This ignores the extended shape of the wheel, even though the motion of different parts of the rim is exactly what matters in rotational analysis.

B. A train traveling 120 km between two cities can be treated as a point mass if we only care about its change in position during the trip.
This is a valid use of the point-particle approximation because the size of the train is negligible compared with the travel distance. In kinematics, when $L_{\text{object}} \ll L_{\text{motion}}$, we can model the object by its center of mass and use a single position vector $\vec{r}$ to describe the motion.

C. We should never model motion in a vacuum, even when we intentionally want to neglect air resistance.
This confuses a physical model with literal reality; assuming a vacuum is a standard idealization used precisely to remove drag from the analysis.

D. None of the above statements is appropriate in physics.
This is incorrect because option B is a standard and appropriate approximation used in introductory mechanics.

Answer: *B*

Q2. Which of the following unit equations is correct?

A. $1\,\text{mg} = 10^{-3}\,\text{kg}$
This is off by a factor of 1000; a milligram is much smaller than a gram, so converting directly to kilograms requires 10^{-6}, not 10^{-3}.

B. $1\,\text{g} = 10^{-3}\,\text{kg}$
The SI prefix relation is $1\,\text{kg} = 1000\,\text{g}$, so dividing both sides by 1000 gives $1\,\text{g} = 10^{-3}\,\text{kg}$. This option has both the correct numerical factor and consistent mass units.

C. $1000\,\text{g} = 10^{-6}\,\text{kg}$
This reverses the metric conversion badly; 1000 g is actually 1 kg, not a millionth of a kilogram.

D. $1\,\text{kg} = 100\,\text{g}$
This uses the wrong power of ten and treats kilo as 10^2 instead of 10^3.

Answer: *B*

Q3. Which option lists an incorrect SI unit for the quantity named?

A. Momentum: $\text{kg} \cdot \text{m/s}$
This is a correct SI unit for linear momentum, so selecting it would confuse momentum with some other quantity.

B. Pressure: N/m^2
This is correct because $1\,\text{Pa} = 1\,\text{N/m}^2$; marking it wrong would ignore the definition of the pascal.

C. Power: $\text{kg} \cdot \text{m/s}^2$
Power is the rate of energy transfer, so its SI unit is $\text{J/s} = \text{W}$, not $\text{kg} \cdot \text{m/s}^2$. The listed unit is actually a newton, since $1\,\text{N} = 1\,\text{kg} \cdot \text{m/s}^2$, so this option mismatches the physical quantity.

D. Energy: $\text{N} \cdot \text{m}$
This is dimensionally correct because $1\,\text{J} = 1\,\text{N} \cdot \text{m}$, so it is not the incorrect choice.

Answer: *C*

Q4. A rescue beacon is located at position $\vec{r} = -5\hat{i} + 12\hat{j}$, where distances are measured in kilometers. Which statement is correct?

A. The beacon is 7 km from the origin.
This comes from subtracting or otherwise combining components incorrectly instead of finding the vector magnitude.

B. The beacon is 13 km from the origin.
The distance from the origin is the magnitude of the position vector, $r = \sqrt{(-5)^2 + 12^2}$. Evaluating gives $r = \sqrt{25 + 144} = \sqrt{169} = 13\,\text{km}$, so this statement is correct.

C. The beacon is 17 km from the origin.
This adds the component magnitudes, $5 + 12$, rather than using the Pythagorean relation for the magnitude of a two-dimensional vector.

D. The beacon is 12 km from the origin.
This mistakes the y-component for the full distance from the origin.

Answer: *B*

Q5. If the displacement of an object is given by $\vec{s} = [(0.80\,\text{m/s}^3)t^3 - (1.60\,\text{m/s}^2)t^2]\,\hat{i}$, which statement is correct?

A. This motion has constant acceleration.
Differentiating twice produces an acceleration that depends on t, so this choice incorrectly treats a cubic position function as constant-acceleration motion.

B. The magnitude of the acceleration at $t = 0$ is $3.2\,\text{m/s}^2$.
From $s_x = 0.80t^3 - 1.60t^2$, the velocity is $v_x = 2.40t^2 - 3.20t$ and the acceleration is $a_x = 4.80t - 3.20$. At $t = 0$, $a_x = -3.20\,\text{m/s}^2$, so the magnitude is $3.2\,\text{m/s}^2$.

C. The magnitude of the acceleration is always $1.6\,\text{m/s}^2$.
This uses the coefficient of the t^2 term directly and skips the two derivatives needed to obtain acceleration.

D. The magnitude of the acceleration at $t = 0$ is $0\,\text{m/s}^2$.
This likely comes from overlooking the constant term that appears after differentiating the displacement function twice.

Answer: *B*

Q6. An object takes 4.0 s to move from position $\vec{r}_A = 1\hat{i} - 7\hat{j}$ to position $\vec{r}_B = 9\hat{i} - 1\hat{j}$ along a straight path, where distances are measured in meters. Which statement is correct?

A. The moving distance is 6.0 m.
This uses only the change in the y-coordinate and ignores the simultaneous change in the x-coordinate.

B. The speed at $t = 0$ must be 2.5 m/s.
This treats average speed, distance divided by time, as if it were guaranteed to equal the initial speed; the initial speed is not determined by the information given.

C. The moving distance is 10.0 m.
The displacement is $\Delta\vec{r} = (9 - 1)\hat{i} + [(-1) - (-7)]\hat{j} = 8\hat{i} + 6\hat{j}$. Because the path is straight from A to B, the distance traveled is $|\Delta\vec{r}| = \sqrt{8^2 + 6^2} = 10.0\,\text{m}$.

D. The acceleration at $t = 0$ must be $0.63\,\text{m/s}^2$.
This invents a specific acceleration from distance and time alone, but many different velocity-time histories could connect the same two positions in 4.0 s.

Answer: *C*

Q7. A soccer ball is kicked with initial speed v_0 at an angle θ above the horizontal. Neglecting air resistance, what is the speed of the ball at its maximum height?

A. 0
This confuses zero vertical velocity at the top with zero total velocity; the ball still has horizontal motion.

B. $v_0 \sin\theta$
This is the initial vertical component of velocity, not the speed at the top of the trajectory.

C. v_0
This ignores that gravity changes the vertical component of the velocity during flight, so the speed is not generally unchanged at the top.

D. $v_0 \cos\theta$
At maximum height, the vertical component of velocity is zero, $v_y = 0$, while the horizontal component remains constant if air resistance is neglected. Therefore the speed there equals $|v_x| = v_0 \cos\theta$.

Answer: *D*

Q8. A drone is moving with initial velocity $\vec{v}_0 = (15.0\,\text{m/s})\hat{j}$ when its thrusters turn on, producing an acceleration $\vec{a} = (-1.5\,\text{m/s}^2)\hat{i} + (2.0\,\text{m/s}^2)\hat{j}$. Which statement is correct?

A. The x-component of the drone's velocity remains zero.
Because $a_x = -1.5\,\text{m/s}^2 \neq 0$, the x-component of velocity begins changing immediately from its initial value.

B. The y-component of the drone's velocity remains constant.
A nonzero y-acceleration means v_y changes with time, so this choice ignores the effect of $a_y = 2.0\,\text{m/s}^2$.

C. Both the direction of motion and the speed change.
The velocity changes according to $\vec{v}(t) = \vec{v}_0 + \vec{a}t$. Since $\vec{a}$ has a component perpendicular to the initial motion, the direction changes, and since it also has a component along the motion, the magnitude of $\vec{v}$ changes as well.

D. Neither the speed nor the direction changes.
This would require zero acceleration, but the thrusters provide nonzero acceleration in both coordinate directions.

Answer: *C*

Q9. If the magnitudes of both charges are doubled and their separation is cut in half, what happens to the magnitude of the electrostatic force between them?

A. It stays the same.
This ignores that both the numerator and denominator of Coulomb's law change, and in this case they do not cancel.

B. It decreases by a factor of 4.
This treats the increased charge as if it weakened the force or mishandles the inverse-square dependence on distance.

C. It increases by a factor of 8.
This captures only part of the scaling and misses that halving r changes r^2 by a factor of $1/4$.

D. It increases by a factor of 16.
Coulomb's law gives $F = k\dfrac{|q_1 q_2|}{r^2}$. Doubling both charges multiplies the numerator by 4, and halving the separation makes the denominator $\left(\frac{1}{2}r\right)^2 = \frac{1}{4}r^2$, which multiplies the force by another factor of 4, so the net change is $4 \times 4 = 16$.

Answer: *D*

Q10. An object can escape a planet of mass M and radius R if its speed is at least $v_e = \sqrt{2GM/R}$. Based on this expression, which statement is correct?

A. A 500 kg probe needs a higher escape speed than a 5 kg probe launched from the same planet.
This confuses the mass of the escaping object with the planet's mass; the formula contains M and R, not the probe mass.

B. For a fixed planet mass, increasing the planet's radius lowers the escape speed.
From $v_e = \sqrt{2GM/R}$, we have $v_e \propto 1/\sqrt{R}$ when M is constant. So making the planet larger in radius decreases the required escape speed because the same mass is spread farther from the launch point.

C. Decreasing the planet's mass raises the escape speed.
This reverses the dependence on M; since $v_e \propto \sqrt{M}$, a smaller planet mass leads to a smaller escape speed.

D. A planet with greater mass and greater radius must always have a lower escape speed.
This claims a definite outcome even though increasing M and increasing R have competing effects, so the result depends on their relative changes.

Answer: *B*

Q11. A constant force $\vec{F} = (6.0\,\text{N})\hat{i} + (-8.0\,\text{N})\hat{j}$ acts on a particle as it moves along the y-axis from $y = -1.5\,\text{m}$ to $y = 2.0\,\text{m}$. What is the work done by the force?

A. -28.0 J
Work is the dot product $W = \vec{F} \cdot \Delta\vec{r}$. Here the displacement is purely vertical, $\Delta\vec{r} = (0)\hat{i} + (3.5\,\text{m})\hat{j}$, so $W = F_y \Delta y = (-8.0)(3.5) = -28.0\,\text{J}$.

B. 21.0 J
This uses the x-component of the force even though there is no x-displacement, so that component does no work.

C. 35.0 J
This is close to multiplying the force magnitude by the distance, which is incorrect because work depends on the component parallel to the displacement.

D. 6.0 J
This likely takes only the x-component or mishandles the displacement interval and does not apply the dot-product definition of work.

Answer: *A*

Q12. A 0.80 kg rubber ball is moving at 12 m/s toward a wall. It rebounds straight back at 9.0 m/s, and the contact lasts 0.050 s. What is the magnitude of the average force on the ball during the collision?

A. 48 N
This uses an incomplete change in velocity and ignores that the rebound reverses the sign of the final velocity.

B. 336 N
Take motion toward the wall as positive, so $v_i = +12\,\text{m/s}$ and $v_f = -9.0\,\text{m/s}$. Then $F_{\text{avg}} = m\Delta v/\Delta t = 0.80\,(-21)/0.050 = -336\,\text{N}$, so the magnitude of the average force is 336 N.

C. 168 N
This commonly comes from using only one of the speeds or forgetting that reversing direction makes the velocity change larger than $12 - 9$.

D. 16.8 N
This underestimates the impulse dramatically and likely results from dividing by time incorrectly.

Answer: *B*

Q13. A 62 kg skateboarder is initially at rest. She catches a 2.5 kg medicine ball thrown horizontally at 8.0 m/s. What is her speed immediately after catching the ball?

A. 0.12 m/s
This is too small because it underestimates the incoming momentum or overestimates the combined mass effect.

B. 0.31 m/s
Because the ball is caught, this is a perfectly inelastic collision and linear momentum is conserved: $m_b v_b = (M + m_b)v_f$. Substituting $m_b = 2.5\,\text{kg}$, $v_b = 8.0\,\text{m/s}$, and $M = 62\,\text{kg}$ gives $v_f = \dfrac{2.5 \times 8.0}{64.5} \approx 0.31\,\text{m/s}$.

C. 8.0 m/s
This assumes the skateboarder moves off with the ball's original speed, ignoring conservation of momentum and the much larger total mass after the catch.

D. 3.2 m/s
This is far too large for a light projectile being caught by a much more massive person and reflects a momentum-calculation error.

Answer: *B*